PQC Migration on Public Chains — Block 15

Resource estimates into judging criteria · crypto agility as a governance property · the Bitcoin lane in detail

Blockchain Governance Initiative Network

BGIN – Mitchell Travers & Shin'ichiro Matsuo · IKP Working Group

BGIN Block #15 | Georgetown University Capitol Campus, Washington, D.C. | 15 October 2026, Room B, 09:50–13:00 | hybrid

Workshop W2 of the PQC Migration Prize series · continues the GDC26 breakout (Geneva, 2–3 Sep 2026) · ecosystem-neutral: no chain and no scheme is endorsed

The Morning in One Page

The keynote sets the frame; the two IKP sessions do the drafting. This is Workshop W2: the evaluation committee is seated here, the metric register moves v0 → v1, and the scheme-scope rulings are argued.

Time	Slot	What it is for
09:20–09:50	Keynote (Room A)	*Crypto Agility in the Quantum and AI Era* — Alex Pruden, Project Eleven
09:50–11:20	IKP: PQC Migration (1)	Resource estimates and the ECDSA.fail community's numbers into judging criteria — **with the Bitcoin lane in detail**
11:30–13:00	IKP: PQC Migration (2)	Crypto agility as a governance property; the NIST signature migration path, with NIST participation; the W2 decisions

🔗 **Takeaway:** Leave with a seated committee, a v1 metric register, scheme rulings on record, and named owners for the open questions — not a locked migration spec.

The Job Is Not to Pick Another Primitive

First PQC standards are in place (ML-KEM, ML-DSA, SLH-DSA; FN-DSA in draft; SP 800-208 stateful hash-based). Re-running a competition in this room would waste the morning.

What still has no agreed playbook for public ledgers, wallets, and credentials:

- How to stage a hybrid and then a cutover when every node must agree on validity rules.
- How to keep verifying old signatures for as long as funds and archives remain live.
- How to rotate keys and addresses without stranding dormant accounts.
- How to tell a genuine PQ-ready stack from a screenshot.

🔗 **Takeaway:** Treat PQC as a multi-year systems and governance program, not a weekend library swap. GDC26 named the gaps; Block 15 writes the criteria.

From Geneva to Washington — What Moved Since GDC26

Named at GDC26 (Sep 2026)

- DLT-specific deployment gaps: activation, wallets, L2, dormant funds, evidence
- Resource estimates and migration design in one conversation
- PQ ZKP and MPC as key-management and proof-path changes
- Japan's PQC Migration Prize as one measurement path, not a second standard

On record since (public, draft v0)

- Coordination repository chartered: outcomes O1–O6, governance, metric register M-01..M-22, scheme register, testbed and harness spec, roadmap W1–W6
- Public Discourse category open; nominations open; conveners named at W1
- Open-benchmark model for calibration tracks proposed (decision 0001) with a first track, T1 *verify-single*
- SSR 2026 PQC workshop (13 Dec, Baltimore) = W4 anchor; papers due 12 Oct

🔗 **Takeaway:** Nothing is final until the committee ratifies it and the host publishes the call. Today is where the committee starts existing.

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Session 1 · 09:50–11:20

Resource Estimates into Judging Criteria

What would actually be required to break secp256k1, under which assumptions, with which error bars — and how that becomes a scored line in a rubric rather than a slogan.

The Numbers on the Table

Migration urgency is a function of resource estimates. This room holds the estimate and the architecture in one place; policy rooms usually hear them from different speakers.

Source	What it says
Google Quantum AI et al., *PRX Quantum* 7, 031001 (Aug 2026; arXiv 2603.28846)	secp256k1 ECDLP at about 1,200 logical qubits and 90M Toffoli gates, or 1,450 / 70M; attacks sorted **on-spend** / **at-rest** / **on-setup**; mitigations ranked by on-chain consequence; ZK-substantiated estimates without published attack circuits
ECDSA.fail community benchmark	The continuing open resource-estimation benchmark: solvers and agents under public notes, re-run by a validator, promoted by measured improvement
Project Eleven, *Key recovery from state reuse* (Sep 2026)	The stateful hash-based failure is demonstrated, not theoretical: signing twice from one LMS / XMSS state exposes the key

🔗 **Takeaway:** The estimates set the clock. No calendar marketing of a single Q-day — threat narratives are scenarios, and the register scores which keys are already public.

On-Spend, At-Rest, On-Setup — Timing Is Not One Number

On-spend

The public key is revealed only when funds move. The race is between the legitimate owner and an attacker who sees the broadcast — a fast-clock CRQC during confirmation, not an unbounded offline grind.

At-rest

The public key is already on-chain. Any CRQC, at any later date, can grind offline against the stored verification key and forge a spend.

On-setup

A universal classical exploit manufactured once by a one-off quantum computation, before any attack — the case that breaks the “we will see it coming” assumption.

Address design, key reuse, and wallet defaults therefore change the threat model. Metric open question 6 asks, for each of the 22 metrics, which of the three classes it defends against.

🔗 **Takeaway:** Resource estimates only become policy-useful when they are tied to which keys are already public. That is a Bitcoin-specific question — next section.

How an Estimate Becomes a Criterion

The register scores substantiation, not fear. Three rows carry the resource-estimate work; the rest are performance and agility.

- **M-19 Security review score** — checklist: assumptions, parameter set, side channels, stateful-key handling; seeds from the PQC Audit Index six sections, to be adopted at W2.
- **M-22 PQC-readiness claim substantiation** — checklist against the *"PQC-ready chain" definition*, the December (W4) deliverable; scores a stack against a published definition rather than a claim.
- **M-20 Governance fit** — transition-period design, dormant / vulnerable asset treatment, the owner-versus-attacker race: a structured narrative, discussion input, not a policy decree.
- **Triage axes** still needing a first cut: protection lifetime · impact if broken · migration cost and dependencies.
- **Gate G1** — the primitive is NIST-standardized or NIST-selected, or a committee-ruled hybrid / stateful hash-based scheme. No new primitives, ever.

🔗 **Takeaway:** An estimate enters the rubric as a declared assumption with error bars, scored under M-19 and M-22 — never as a date.

Session 1 — Discussion Items

1. What would make a **resource estimate** usable for an evaluator — assumptions, error bars, and what must not be collapsed into a Q-day date? What language does the rubric refuse (single Q-day, vendor screenshots)?
2. Which **objects** move first: at-rest keys, on-spend paths, credentials, the proof and key-management path (PQ ZKP, MPC)? Open question 6 needs an owner.
3. Should the ECDSA.fail benchmark's **public-note discipline** (exact model and harness named, re-run by a validator) be the admission rule for agent-produced numbers in the record? (Decision 0001.)
4. What does the **"PQC-ready chain" definition** have to contain for M-22 to be scorable by December?

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The Bitcoin Lane · inside Session 1

Bitcoin in Detail: Exposure, Cost, Paths, and What Gets Measured

Bitcoin is one of the two evidence anchors. This section is for the Bitcoin Core and Bitcoin infrastructure engineers in the room: what the register asks of Bitcoin evidence, and what it deliberately leaves alone.

Why Bitcoin Does Not Migrate Like TLS

Enterprise PKI / TLS	Bitcoin
An operator can force a cutover	No single party can force a soft fork; users, miners, node operators and implementations each hold a practical veto
Cipher suites can be negotiated per session	Script validity rules are global and sticky; a new output type needs activation
Certificates expire on a known clock	On-chain signatures must stay verifiable for decades; the UTXO set is the archive
Harvest-now, decrypt-later is mainly archival	Public keys sit in the clear in P2PK outputs, reused addresses, Taproot outputs and every spend
Evidence lives in a vendor report	"PQ-ready" claims need comparable measurement: vbytes and weight per block, validation time, UTXO growth

🔗 **Takeaway:** A TLS playbook fails on three axes at once: decentralized upgrades, long-lived authenticity, and economic incentives. The register is written for those three.

Bitcoin’s Exposure Classes — Which Keys Are Already Public

The on-spend / at-rest split is not abstract on Bitcoin: it is decided by output type and by wallet behaviour. Both are measurable from the chain.

Output type	Key exposure	Threat class before any spend
P2PK (early coinbase era)	Public key in the output	At-rest — exposed since creation
P2PKH / P2WPKH, never reused	Hash only; key revealed on spend	On-spend — the confirmation window is the race
P2PKH / P2WPKH, address reused	Key revealed by the first spend	At-rest for every later output to that address
P2SH / P2WSH multisig	Script hashed; keys revealed on spend	On-spend, but keys are reused across a wallet’s history
P2TR (Taproot, BIP-341)	Tweaked x-only key in the output	At-rest — by design; the key-path is the point
Lightning channel outputs	Funding is P2WSH / P2TR; keys revealed on close	Mixed; channel lifetime is the exposure window

🔗 **Takeaway:** Any Bitcoin migration design has to say what it does for each row. The register does not decide the fate of exposed coins; M-20 records the positions.

What the Candidates Cost in Bitcoin Units

Baselines: B-01 secp256k1 ECDSA (legacy / SegWit v0) and B-02 Schnorr, BIP-340 (Taproot) — 33 or 32-byte keys, 64–72-byte signatures. Everything below is scored as a delta against those.

Row	Scheme / parameter set	pk bytes	sig bytes	Note for the Bitcoin lane
B-02	Schnorr BIP-340	32	64	baseline; witness discount applies
S-03	FN-DSA / Falcon-512 (FIPS 206 draft)	897	about 666	smallest lattice signatures; float signing is the implementation risk
S-01	ML-DSA-44 (FIPS 204)	1,312	2,420	the briefing's starting point
S-05	LMS SHA256 M32 H10 (SP 800-208)	60	1,456	stateful; state reuse is a demonstrated key-recovery
S-04	XMSS-SHA2_10_256 (SP 800-208)	64	2,500	stateful; same caveat
S-02	SLH-DSA SHAKE-128s (FIPS 205)	32	7,856	conservative assumptions; large

Sizes from testbed/SCHEMES.md. Witness bytes count one weight unit each: a signature 10–120× Schnorr's size is 10–120× the witness weight; M-07 turns that into vbytes per reference block mix.

🔗 **Takeaway:** Size is the first-order Bitcoin cost, not the only one: verify time, propagation and UTXO growth are measured, not assumed.

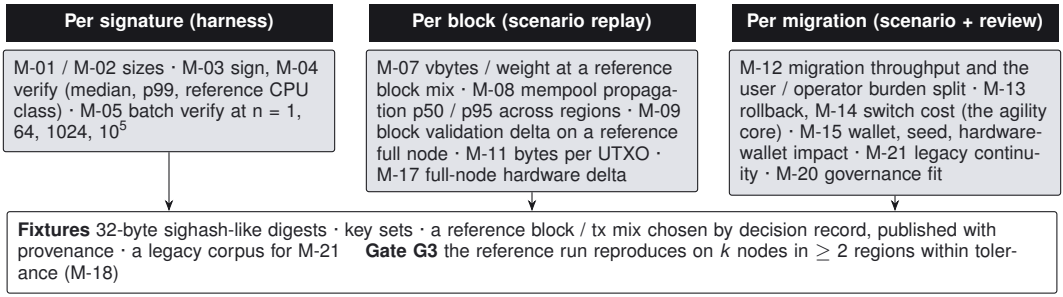
Paths Under Ruling — What Counts as In Scope for Bitcoin

The scheme-scope rulings are a formal vote of the committee, recorded in decisions/. Two of the five composition rows are Bitcoin-facing.

- **C-01 Hybrid classical + PQ** — Schnorr or ECDSA alongside ML-DSA, in the IETF LAMPS composite direction. The GDC26 recommendation is “hybrid then cutover”; the harness measures the composite as one unit. *Ruling needed: in scope as a path?*
- **C-02 Quantum-resistant output type** — proposals in the P2QRH / BIP-360 direction. Proposal-stage; a **reference for the BTC mapping only, no endorsement**. *Ruling needed: reference mapping, or out of the register until a BIP number is assigned?*
- **S-04** / **S-05 stateful hash-based** — in scope with state handling as a *scored* property, not a footnote; whether non-SP-800-208 parameterizations count is a separate ruling.
- **Legacy continuity (M-21)** — pre-migration signatures must still verify after cutover; a pass / fail gate plus node cost. For Bitcoin this is the UTXO set, not an archive.

🔗 **Takeaway:** BGIN rules on what the *testbed measures*, never on what Bitcoin activates. The rulings make Bitcoin evidence comparable; they do not make policy.

What the Testbed Measures on the Bitcoin Lane



🔗 **Takeaway:** Every performance number is a ratio against B-01 / B-02 on the same node. A Bitcoin reviewer gets deltas in the units they already use.

What Stays Outside Bitcoin Core's Remit — and Outside BGIN's

Not asked of Bitcoin Core

- Adopting, endorsing or reviewing any scheme, output type or BIP because it appears in the register
- Deciding the fate of exposed-key or dormant coins
- Running or hosting the testbed — node hosts are universities and neutral labs, funded transparently
- Sitting on the committee as "Bitcoin" — seats are individuals with COI disclosure

Not done by BGIN

- Chain endorsement, vendor scorecards, or a Q-day date
- New primitives or parameter sets — NIST is the reference
- Prize adjudication — the host's process; BGIN organizes the committee and testbed
- Policy decrees on transition design — O4 is discussion input, recorded, not ruled

🔗 **Takeaway:** The register asks Bitcoin engineers for one thing: that the Bitcoin deltas are measured the way a Core reviewer would measure them. That needs Bitcoin engineers writing the fixtures.

Three Asks to the Bitcoin Engineers in the Room

1. **Technical-advisor seat, Bitcoin track** — input on baselines, the reference block / tx mix, the M-15 wallet checklist, without winner-selection votes. COI form; nominations are open on the Discourse category.
2. **Fixtures by pull request** — the sighash-like corpus, the reference block mix and its provenance, the legacy corpus for M-21. The repository is public and every commit is signed and signed-off; a PR is the record.
3. **A node, or a name** — the node-host call opens at W3 (November): a reference full node in a university or neutral lab, in a region the testbed lacks. Or the name of the lab that should be asked.

The committee's Bitcoin member seats (2–3, per the proposed composition) are the primary evaluation focus for the BTC-tx mapping. They are individuals, not projects.

🔗 **Takeaway:** Bitcoin's evidence is in the standard only if Bitcoin engineers put it there. Nothing in the process can substitute for that.

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Session 2 · 11:30–13:00

Crypto Agility as a Governance Property; the NIST Migration Path

Agility is harder than readiness and is what future-proofs the system. This session writes the agility and governance rows, hears the NIST signature migration path, and takes the W2 decisions.

Agility Is Harder Than Readiness — Outcome O3

Evidence on the **mechanism of switching** — hybrid and staged migration, scheme substitution cost, rollback, policy flexibility — because the next migration is already certain and the primitive list will move again.

Metric	What	Unit	How measured
M-14	Agility switch cost	time + coordination steps	scenario: swap scheme A to B under the proposed mechanism — the core agility metric
M-13	Rollback cost	time + coordination steps	scenario: revert a staged migration
M-12	Migration throughput	keys / accounts per unit time	scenario; documents the user vs node-operator burden split
M-21	Legacy verification continuity	pass / fail + node cost	pre-migration signatures still verify after cutover

🔗 **Takeaway:** Design for the next migration now. A rushed cutover creates new incidents; a frozen profile goes stale. Both are scored.

Crypto Agility as a Governance Property — Outcome O4

Who upgrades first

Node operators or users? M-12 makes the burden split explicit. On Bitcoin the answer differs by output type; on Ethereum by account model.

The migration window

The legitimate-owner-versus-attacker race is a governance question with a measurable shape: window length, what a wallet default does, what a rollback costs.

Dormant and vulnerable assets

Treatment of exposed-key and dormant funds is recorded as positions, scored under M-20 as a structured narrative — discussion input, never a decree.

"PQC-ready chain" — the definition M-22 scores against — is the December deliverable (W4, SSR workshop). It has to be written so that ST/PP and wallet profiles can rotate algorithms without a fork-by-fork rewrite.

🔗 **Takeaway:** Agility written as a governance property is what lets the same rubric travel: competition → BGIN Standard → ISO/TC 307 → Japan's crypto-asset security criteria.

The NIST Signature Migration Path — What the Register Leans On

NIST is the algorithm and parameter authority for everything in scope; it is not a competition lane. Measurement-methodology cooperation is sought where appropriate.

- **FIPS 204 ML-DSA**, **FIPS 205 SLH-DSA** — standardized; S-01 and S-02.
- **FIPS 206 FN-DSA (Falcon)** — draft; final expected late 2026 or 2027; S-03 is confirmed at the scope ruling.
- **SP 800-208** XMSS and LMS / HSS — stateful hash-based; S-04 and S-05 with state handling scored.
- **ACVP** / **CAVP** wording is the T1 correctness gate for calibration tracks.
- Hybrid / composite signatures follow the **IETF LAMPS** direction (C-01).
- The on-ramp round-3 signature candidates are **out of scope by gate G1**.

🔗 **Takeaway:** The register tracks NIST's list; it never extends it. When the list moves, the agility metrics are what say whether a stack can follow.

What Block 15 Decides — the W2 Outputs

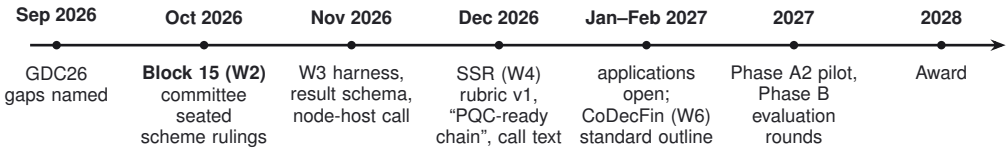
Decision	Method	Lands as
Committee seated (Phase A); chairs elected	nominations + COI forms; formal vote	COMMITTEE.md roster
OUTCOMES O1–O6 ratified (if not at W1)	formal vote, quorum of half	decisions/ record
Metric register v0 → v1; directions per metric	rough consensus on the PR	evaluation/METRICS.md
Scheme-scope rulings: C-01 hybrid, C-02 P2QRH reference, S-04 non-SP-800-208 parameters, C-03 / C-04 / C-05	formal vote per ruling	decisions/ records
Open-benchmark model for calibration tracks	vote on decision 0001; recusals recorded	decisions/0001 accepted
Open questions 1–5 assigned owners	conveners call it	workshop W2 record

🔗 **Takeaway:** Discuss on Discourse, record by pull request, rule by decision record. Every vote and every scope ruling gets an ADR.

The Open Questions That Need Owners Today

1. Is the headline quantity **total overhead** or **migration speed**? The rubric decides the weights.
2. Who must upgrade first — **node operators or users** — and how does M-12 reflect the burden split?
3. How to score the **owner-versus-attacker race** during a migration window (M-20, track a)?
4. Which **reference CPU class** and **reference tx mix** / **block** anchor M-03 to M-09? Decision record 0002 territory, with k and the tolerance band.
5. Are **consensus-layer** (ETH-cons) and **user-level** (BTC-tx, ETH-user) evidence scored on one rubric or two?
6. For each metric, which attack class — **on-spend, at-rest, on-setup** — does it defend against?

From This Room to the 2028 Award



Competition / evaluation → BGIN Standard → ISO/TC 307 → Japan’s crypto-asset security criteria

Dates are indicative until the host confirms the call. Between workshops: Discourse carries the discussion, PRs carry the record, decision records carry the rulings.

🔗 **Takeaway:** This morning can still change what the call asks teams to prove. SSR on 13 December is the next drafting stop.

How to Engage

- **Evaluation committee** — draft call text, metrics and rubrics in Phase A; nominate via the issue template.
- **Technical advisor** — Bitcoin / Ethereum / measurement input without winner votes.
- **Testbed node host** — a university or neutral lab; contracting later; role separated from judging.
- **SSR workshop, 13 Dec** — papers due 12 Oct; accepted papers feed the competition specification.
- **Submit** / **build** — when the official call opens (planned 2027).

Repository
github.com/bgin-global/pqc-agility-competition

Discourse category
bgin.discourse.group/c/post-quantum-crypto-agility-competition/44

🔗 **Takeaway:** Discussion on Discourse; the repository is the record. Nothing is final until the committee ratifies it and the host publishes the call.

The hard problem is deployment, evaluation, and migration governance.

Not selecting a new primitive. Not a date. Not an endorsement.

BGIN IKP Working Group · Post-Quantum Crypto Agility Competition category
on the BGIN Discourse · github.com/bgin-global/pqc-agility-competition

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